

Setu: Cancer Treatment Dropout Recovery

Reminders don't bring patients back. Removing their barrier does.

TRACK
Cancer

PRIMARY USER
Doctor / Care Team

USE CASE
Patient Follow-up &
Continuity of Care

NATURE
Assistive · non-clinical

Prototype: hannielvinu.github.io/Setu · Code: github.com/hannielvinu/Setu

1. The problem

Long-term cancer care is a sequence of time-bound steps: chemotherapy cycles, radiotherapy fractions and reviews, and years of surveillance. Patients quietly drop out between these steps.

34%

of cervical cancer patients who completed treatment were lost to follow-up (South India, n=690); 12% did not complete planned treatment [1]

19–50%

treatment abandonment reported for paediatric AML across Indian centres [2]

>100 km

distance from the hospital: a consistently reported risk factor for loss to follow-up [1]

How it is handled today. Coordinators and patient navigators (e.g. TMC's KEVAT programme [4]) search registers, spreadsheets and appointment lists by hand and cold-call patients. This works but does not scale, and **the reason a patient dropped out is rarely recorded**, so the cause is never fixed. Integrating with hospital software is costly (₹30–50 lakh per mid-sized facility for ABDM-compliant integration [3]), so any solution that needs EMR integration will not reach most hospitals.

2. The insight

Many dropouts are driven by **non-clinical barriers**: scheme (PM-JAY / state) paperwork, travel and lodging costs, family and caregiver duties, "I feel better", fear of side effects, or simply not knowing the next date. SMS reminders remove none of these. Patient apps miss the patients who have already gone silent. Human navigators help, but one person cannot reach every patient.

Gap Setu fills: a hospital-run, assistive outreach loop that finds silent dropouts, learns *why* in the patient's own language, removes the barrier where it can, hands everything clinical to people, and gives the hospital its first data on why patients leave.

3. The solution: a closed loop

1 · Find

Clinician-defined pathway templates and simple rules flag overdue patients from existing exports or clerk-verified register photos.

2 · Reach

WhatsApp/SMS heads-up, then a consented voice call in the patient's language. Identity check before any disclosure.

3 · Understand

One question: "what is stopping you from coming?" The AI sorts the answer into a fixed barrier list.

4 · Resolve

Rebook a vacant slot, send the scheme document checklist, share lodging help, copy the caregiver, or hand over to a person.

5 · Measure

Return-to-care tracking and a barrier dashboard so the hospital can fix root causes.

Illustrative journey (synthetic)

Shanti, 52, chemo cycle 4 of 6, 9 days overdue, lives 140 km away. She is ranked #2 on the morning worklist (9 days × clinician tier 3 × 1.5 for one past miss). A WhatsApp heads-up arrives from the hospital care-desk number with a neutral sender name. A Hindi voice call follows; her son, a registered caregiver, confirms her birth year. Barrier captured: *"Ayushman card lapsed and the bus fare is too much."* Setu sends the scheme document checklist and lodging list, books Thursday's cancelled day-care slot, and the coordinator sees a one-line summary. All of this runs end to end in the prototype.

4. Scope compliance: assistive by design

OUT-OF-SCOPE ITEM	HOW SETU STAYS OUTSIDE IT
Diagnosis	Setu never sees or infers a diagnosis. It works only from schedule facts (pathway, step, last visit date).
Treatment recommendations	Pathways are written by clinicians. Setu only reminds and rebooks what the doctor already planned.
Clinical decision support	The AI's only outputs are fixed non-clinical labels (money, paperwork, travel, lodging, family, felt better, worried, unaware, other).
Clinical risk scoring	Priority = days overdue × clinician-set tier × (1 + 0.5 × past misses). Operational, transparent, no model.
Interpretation of medical data	OCR extracts six non-clinical fields only: reg. no., name, phone, pathway, visit no., last visit date. No drugs, doses or values.
Autonomous clinical advice	Any symptom goes to a nurse queue verbatim, with no advice. Emergency-card phrases trigger the hospital's own printed instruction, read word for word, plus an on-call alert.

Safeguards built into the design

- **Human-verified data:** a clerk approves every digitised row; date, phone and name sanity checks block errors.
- **Privacy (DPDP):** consent at enrolment, neutral sender, identity check (birth year or card digits) before any detail; a wrong person hears only "please ask them to call back".
- **Fail-safe emergencies:** keyword OR LLM match on the hospital's emergency card triggers the path; ambiguous cases are treated as emergencies.
- **Nurse queue, not triage:** the nurse decides and chooses any leaflet, in about 30 seconds per item.

Telephony realities handled

- Heads-up message 10 minutes before the call; verified business caller ID; DLT-registered SMS.
- Calls only in the patient's preferred time window; at most 3 attempts.
- "Say 'talk to a person' at any time" → warm transfer.
- Fallback chain: call drop → WhatsApp voice notes → SMS + missed-call callback → human worklist.

5. Architecture and technology

Data in (no EMR integration)

Appointment / day-care Excel or CSV exports
Register photos → 6-field OCR → clerk verification
Consent and caregiver details at enrolment

Setu core

FastAPI + PostgreSQL
JSON pathway rule engine (clinician-owned)
LangGraph agent orchestration
Sarvam AI speech-to-text, text-to-speech and LLM (Hindi + regional), restricted to fixed labels
Emergency-card matcher (always on)

Channels and people

Exotel voice and missed-call numbers
WhatsApp Business API (templates, Flows, voice notes)
React: worklist, verification, nurse queue, barrier dashboard

Hosting: India-region, encrypted at rest and in transit, role-based access, audit log, transcript retention limits.

Prototype status (built and deployed)

COMPONENT	IN THE PROTOTYPE TODAY	NEXT (PILOT BUILD)
Pathway engine and priority	Working; clinician tier edits re-rank live; unit-tested	Server-side, multi-hospital config
Outreach agent	Full flow: heads-up, call, identity check, barrier capture, rebooking, fallbacks (EN/HI), simulated in the browser	Sarvam voice + Exotel + WhatsApp API
Reply classifier	Transparent keyword rules (EN/HI/Hinglish), emergency card, exclusions	Sarvam LLM limited to the same labels; rules stay on
Register verification	Side-by-side review, confidence flags, sanity checks, CSV import	Live vision OCR on six fields
Nurse queue and dashboard	Working with synthetic baseline	Live hospital data

All data in the prototype is synthetic; "Arogya Hospital" is fictional.

6. Impact and 60–90 day measurement

Pilot design

About 200 overdue patients at one partner hospital: Setu-supported outreach vs usual care.

Primary outcome: % of overdue patients back in care within 30 days.

Secondary: days to re-engagement; coordinator hours saved; call answer rate and fallback reach; identity-check pass rate; OCR correction rate; nurse-queue review time; emergency alerts and outcomes; barrier breakdown.

Who benefits

- **Doctors:** fewer silent dropouts, no extra screens.
- **Coordinators and nurses:** a ranked worklist and summaries instead of combing registers.
- **Patients and caregivers:** help with the real obstacle, in their language.
- **Hospital leadership:** first data on why patients leave, so causes get fixed.

Scale. Pathways are configuration, not code: chemotherapy, radiotherapy and surveillance first; the same engine runs high-risk pregnancy ANC schedules and diabetes review cycles (a maternal pathway is already configured in the prototype). Designed for rollout across National Cancer Grid institutions.

7. Risks and mitigations

RISK	MITIGATION
OCR misreads handwritten registers	Six fields only, confidence flags, sanity checks, mandatory clerk approval; correction rate tracked
Calls ignored or flagged as spam	Heads-up message, verified caller ID, preferred time windows, WhatsApp/SMS/missed-call fallback
Disclosure to the wrong person	Neutral sender, identity check before any detail, named caregivers from consent
Nurse queue overload	Batched queue with transcript + audio clip; no AI triage; review time measured; emergency path separate
LLM errors or made-up content	LLM only classifies into fixed labels; all speech is pre-approved scripts; deterministic scheduling; red-team suite of 100+ multilingual prompts (target: 0 advice, 0 missed emergency phrases)

8. Roadmap to the finale (28 Nov 2026)

WHEN	MILESTONE
Week 1	Field evidence: real dropout data from our doctor's department; coordinator and patient interviews; the hospital's emergency card
Weeks 2–4	Backend and live channels: pathway service, Sarvam voice, Exotel, WhatsApp templates, consent store
Week 5	Safety: red-team suite, nurse queue hardening; maternal pathway
Week 6	Consented mini-pilot (30–50 patients) with a partner hospital
Finale	Live call demo in an Indian language, fallback chain, barrier dashboard and pilot numbers

Team MedNova is doctor-led: our clinician designs the pathways, scripts and safety rules; our technologists build the agent, channels and dashboards.

References

1. Loss to follow-up among cervical cancer patients, South India. PMC9199505. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9199505>
2. Treatment abandonment in paediatric AML, India. Indian J Med Paediatr Oncol 2019. https://www.ijmpo.org/assets/articles/2019/pdf/ijmpo.ijmpo_84_18.pdf
3. From infrastructure to impact: operationalizing ABDM. HMPI, July 2026. <https://hmpi.org/2026/07/09/from-infrastructure-to-impact-operationalizing-indias-ayushman-bharat-digital-mission/>
4. KEVAT patient navigator programme, Tata Memorial Centre. <https://tmc.gov.in/index.php/kevat-patient-navigator>

Setu is assistive: it does not diagnose, recommend treatment, provide clinical decision support, score clinical risk, interpret medical data or give clinical advice. Team MedNova · Health-a-thon 2026.